ACCIDENT ALERT SYSTEM WITH LOCATION

ABSTRACT

- THE "IOT-BASED ACCIDENT DETECTION AND ALERT SYSTEM" IS DESIGNED TO PROVIDE IMMEDIATE ASSISTANCE TO ACCIDENT VICTIMS BY REDUCING THE TIME TAKEN FOR EMERGENCY SERVICES TO REACH THE SPOT. THE SYSTEM INTEGRATES AN **ACCELEROMETER (ADXL335/345)** TO DETECT SUDDEN IMPACTS OR VEHICLE ROLLOVERS, A **GPS MODULE (NEO-6M)** TO ACQUIRE REAL-TIME GEOGRAPHICAL COORDINATES, AND A **GSM MODULE (SIM800L)** TO TRANSMIT THIS INFORMATION. WHEN AN ACCIDENT IS DETECTED, THE SYSTEM AUTOMATICALLY SENDS AN SMS CONTAINING THE GOOGLE MAPS LOCATION LINK TO PRE-DEFINED EMERGENCY CONTACTS, ENSURING THE "GOLDEN HOUR" OF MEDICAL TREATMENT IS UTILIZED EFFECTIVELY.

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OBJECTIVE

- To design a low –cost , automated system for real – time accident detection.
- To extract precise geographical location (latitude and longitude) using gps technology.
- To ensure immediate communication of the accident spot to emergency rsponders via gsm.
- To minimize the mortality rate caused by delay in emergency medical reporting

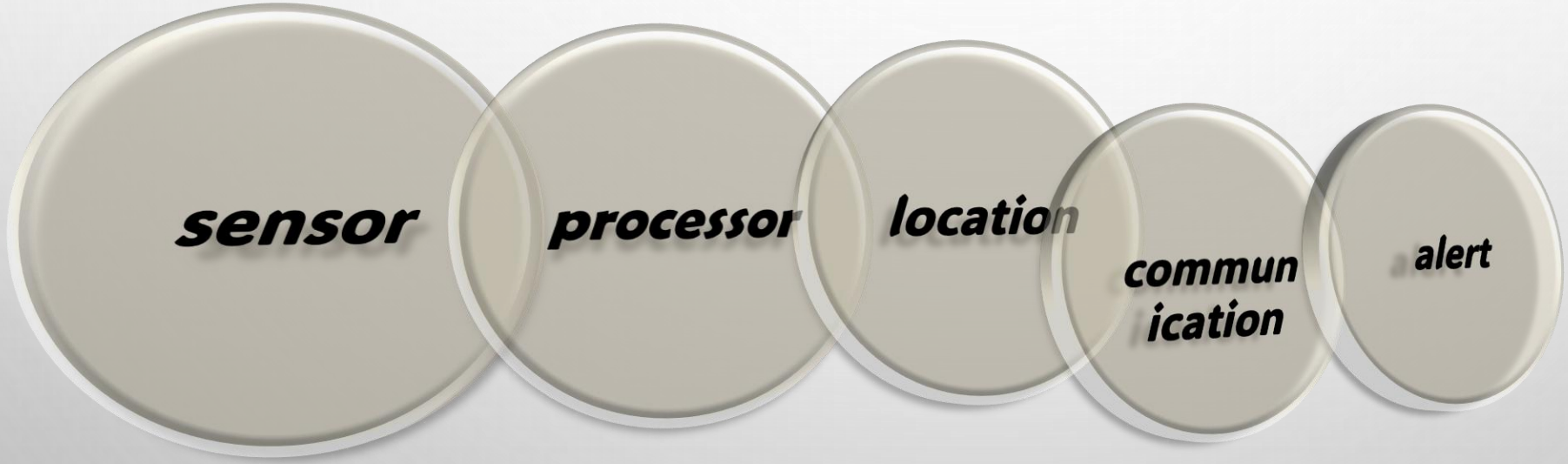
component	function
Arduino nano	Processing & control
ADXL335	Impact & tilt sensing
NEO-6M GPS	Location tracking
SIM800L GSM	SMS alerting
Buzzer	Sound warning

PROBLEM STATEMENT

- ***CURRENTLY, MOST ACCIDENT FATALITIES OCCUR DUE TO A LACK OF IMMEDIATE MEDICAL ATTENTION. IN MANY CASES, ACCIDENTS HAPPEN IN REMOTE AREAS OR DURING LATE HOURS WHERE NO WITNESSES ARE PRESENT TO CALL FOR HELP. EXISTING MANUAL REPORTING SYSTEMS ARE OFTEN TOO SLOW, AND VICTIMS MAY BE UNCONSCIOUS, MAKING THEM UNABLE TO SEEK HELP THEMSELVES. THERE IS A CRITICAL NEED FOR AN AUTOMATED SYSTEM THAT ALERTS AUTHORITIES WITHOUT HUMAN INTERVENTION.***

ARDUINO DIAGRAM

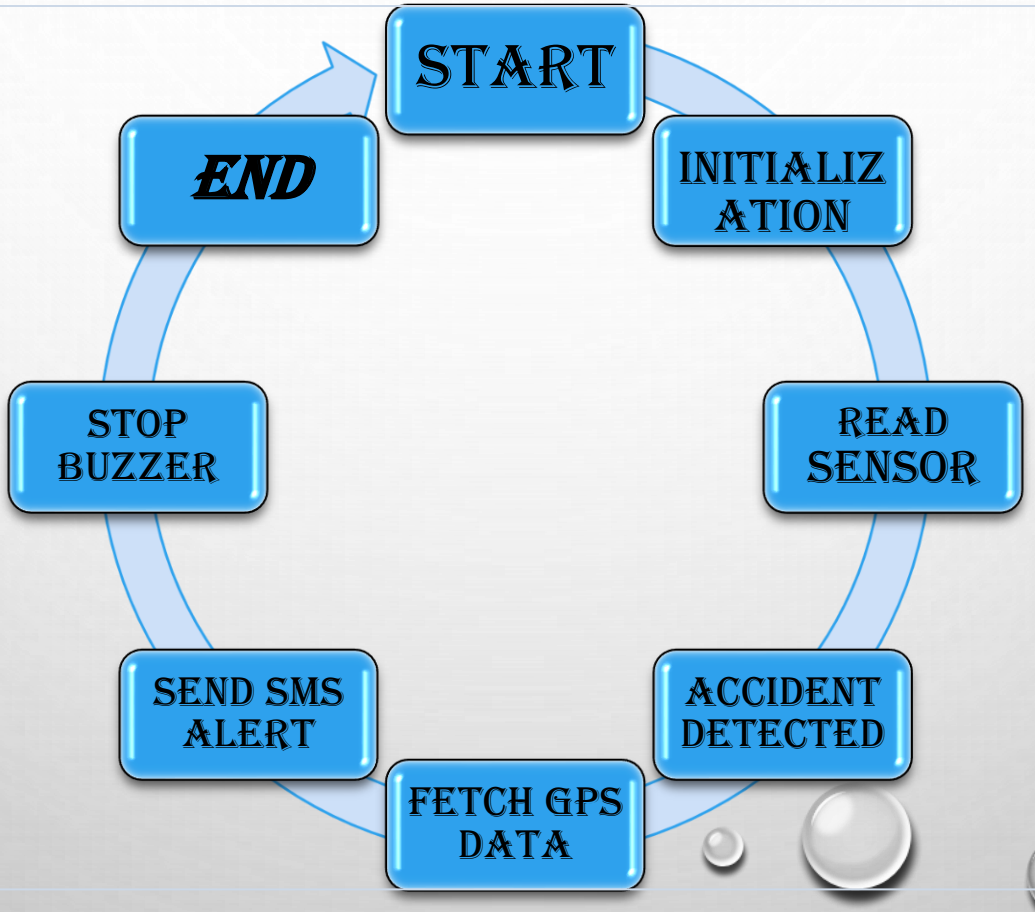
(HARDWARE INTERFACE)



ALGORITHM

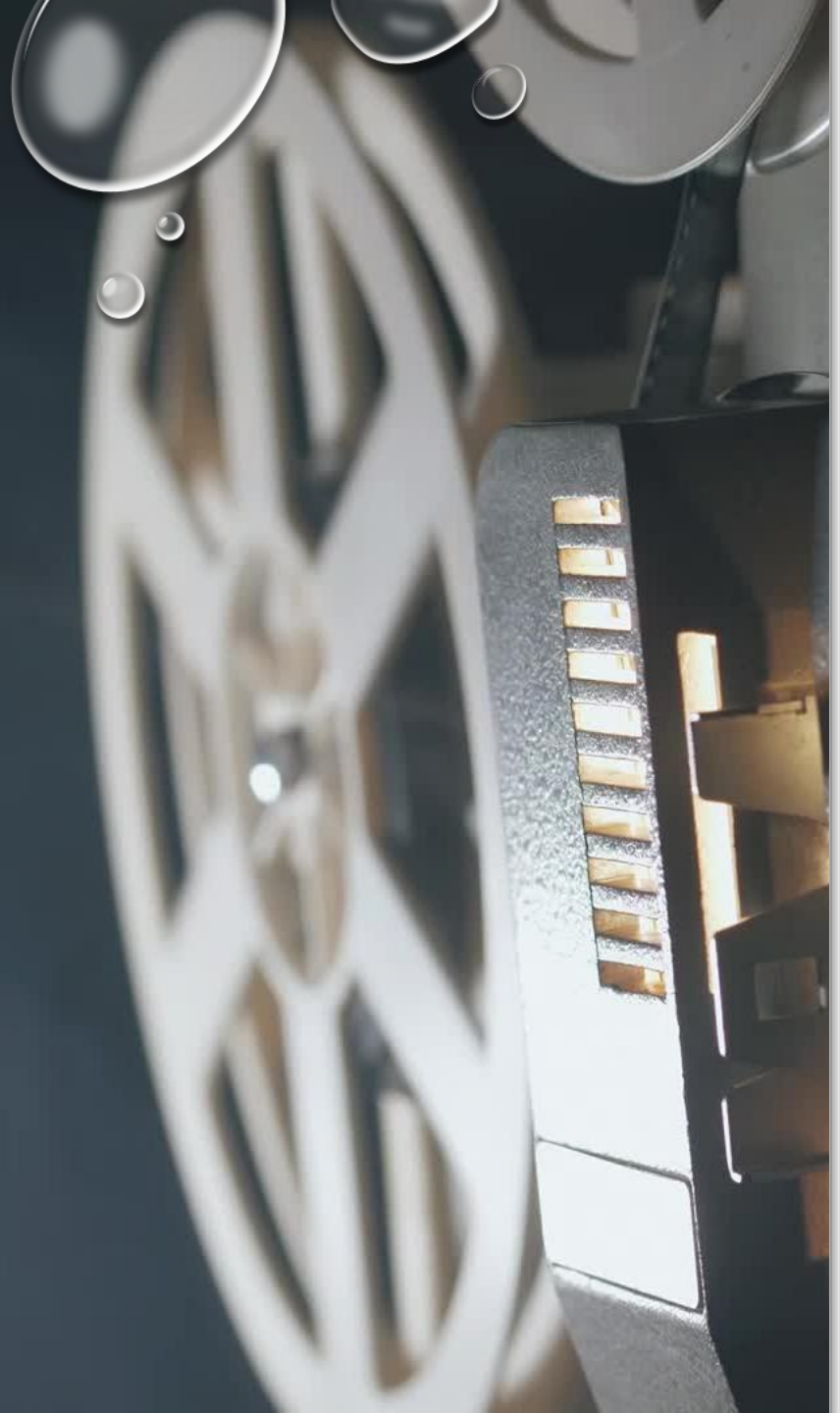
1. *Step 1: Initialize the system, GPS, and GSM modules.*
2. *Step 2: Continuously read values from the Accelerometer (X, Y, and Z axes).*
3. *Step 3: Compare the current values with the preset "Threshold" (limit for impact or tilt).*
4. *Step 4: If the value exceeds the threshold, trigger the buzzer and wait for a 10-second "Grace Period" (to allow the user to cancel if it's a false alarm).*
5. *Step 5: If not canceled, fetch the current Latitude and Longitude from the GPS module.*
6. *Step 6: Format the data into a Google Maps URL.*
7. *Step 7: Command the GSM module to send a text message to the emergency mobile numbers.*
8. *Step 8: Stop the buzzer and return to monitoring*

FLOWCHART


START
INITIALIZ ATION
READ SENSOR
ACCIDENT DETECTED
FETCH GPS DATA
SEND SMS ALERT
STOP BUZZER
END

APPLICATION

- **VEHICLE SAFETY:** CAN BE INSTALLED IN CARS, BIKES, AND TRUCKS FOR PERSONAL AND COMMERCIAL SAFETY.
- **FLEET MANAGEMENT:** LOGISTICS COMPANIES CAN TRACK ACCIDENTS INVOLVING THEIR TRANSPORT VEHICLES.
- **SCHOOL BUSES:** TO ENSURE THE SAFETY OF STUDENTS AND PROVIDE IMMEDIATE ALERTS TO SCHOOL MANAGEMENT.
- **ELDERLY/MEDICAL MONITORING:** CAN BE ADAPTED AS A "FALL DETECTION" SYSTEM FOR ELDERLY PEOPLE LIVING ALONE.



CONCLUSION

- THE DEVELOPED SYSTEM PROVIDES A RELIABLE AND COST-EFFECTIVE SOLUTION FOR EMERGENCY ACCIDENT REPORTING. BY AUTOMATING THE ALERT PROCESS, IT REMOVES THE DEPENDENCY ON BYSTANDERS AND ENSURES THAT HELP IS ON THE WAY WITHIN SECONDS OF AN INCIDENT. FUTURE ENHANCEMENTS COULD INCLUDE INTEGRATING A CAMERA MODULE TO SEND PICTURES OF THE ACCIDENT SPOT OR CONNECTING THE SYSTEM TO A CLOUD PLATFORM FOR CENTRALIZED MONITORING BY CITY TRAFFIC POLICE.